ABIY ABINET MAMO

Computer Science M.S. Candidate | Data Science, ML/AI and Software Engineering
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EDUCATION

M.S. Computer Science, Machine Intelligence and Cognition | Aug 2025 - Present
George Washington University, Washington, DC
B.Tech. Computer Science and Engineering | Sep 2021 - Jul 2025
KIIT University, Bhubaneswar, India

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, Dart
ML and AI: scikit-learn, XGBoost, PyTorch, NLP, LLMs, LangChain, Ollama, MiniLM, model evaluation
Data and Software: pandas, NumPy, FastAPI, MLflow, Qdrant, Docker, SQLite, Flutter, Firebase, Git

RELEVANT EXPERIENCE

Summer Research Intern | IoT Lab, BITS Pilani *May 2023 - Nov 2023*
• Reviewed IoT and machine learning literature; analyzed research papers and contributed practical research ideas.
Data Science Intern | Code Clause *Jun 2023 - Jul 2023*
• Developed a logistic-regression churn baseline and Apriori market basket analysis; built a spaCy resume parser and seller price-recommendation tool.
Teaching Assistant | KIIT University *Sep 2023 - May 2024*
• Designed and delivered data structures instruction in C++ and Python for individual and group learners.

SELECTED PROJECTS

Signal Sprint | React, TypeScript, Cloudflare Workers, D1
GitHub / signal_sprint
• Built a 2-8 player logic game with room codes, synchronized rounds, server-side scoring and persistent state; tested two-player synchronization and six-round scoring and authorization.
RAG Document Intelligence | FastAPI, Qdrant, MiniLM, Ollama
GitHub / rag-document-intelligence
• Built PDF ingestion, persistent Qdrant search and cited answers; MiniLM reached 100% Recall@3 on 12 answerable questions over 6 synthetic documents. Added API tests and optional Ollama generation.
ML Prediction and MLOps Platform | scikit-learn, MLflow, Docker
GitHub / ml-prediction-mlops
• Compared 2 classifiers on 569 records (30 features); validation-selected logistic regression reached 0.996 ROC AUC and 96.5% accuracy on a 114-record holdout. Added MLflow tracking, serving and drift monitoring.
Real-Time Recommendation System | FastAPI, SQLite, item similarity
GitHub / real-time-recommender
• Trained on MovieLens 100K (943 users, 1,682 movies); achieved 6.97% Recall@10 versus 5.66% popularity baseline on 459 relevant holdouts. New ratings personalize the next request without batch retraining.
Multilingual YouTube Summarizer | Python, Llama 3, LangChain, Gradio
GitHub / MultiLingual-YouTube-Summarizer-using-LLAMA3
• Integrated captions, token-based chunking and map-reduce summarization with local Llama 3; exposed summarization controls and 68 configured language choices.
AI Study Planner | Flutter, Dart, Firebase
GitHub / AI-Study-Planner
• Flutter/Firebase client prototype with 6 screen modules and 5 service modules for courses, tasks, authentication and syllabus submission; separate AI backend and build fixes required.

PUBLICATIONS

Advancing Medical Imaging Through Generative Adversarial Networks: A Comprehensive Review and Future Prospects
First author | Cognitive Computation (2024)
Unlocking the Power of Machine Learning in Big Data: A Scoping Survey
Co-author | Data Science and Management (2025)
Decoding the Digital Market: Understanding How Machine Learning Shapes E-Commerce
First author | International Journal of Electronic Business (2026)

AWARDS

Top 1% Award, National Exam of Ethiopia (2021) | Study in India Scholarship (2021 - 2025)